

Discrete Mathematics: Algorithms revision

1 Consider the following algorithm, which takes positive integers as inputs:

- Step 1: Input A
- Step 2: Input B
- Step 3: If $A > B$ then let $A = A - B$
- Step 4: If $B > A$ then let $B = B - A$
- Step 5: If $A = B$ then Output A . STOP
- Step 6: Go to Step 3

- (a) Trace through the algorithm with inputs $A = 68$ and $B = 52$. Show the values of A and B each time Step 6 is encountered, and give the final output.
- (b) How many times is Step 6 encountered when the inputs are $A = 60$ and $B = 40$?
- (c) What is the maximum number of times that Step 6 could be encountered?
- (d) Explain why the instructions above would not comprise an algorithm if the requirement that the inputs are positive were removed.

2 The table below shows the sizes of 10 files. These files need to be put onto USB sticks with a maximum capacity of 16 GB.

File	A	B	C	D	E	F	G	H	I	J
Size (GB)	12.4	8.1	2.3	1.2	7.3	7.4	1.7	8.8	7.5	3.2

Show the packing that is produced when the following algorithms are used:

- (a) next-fit
- (b) first-fit
- (c) first-fit decreasing.

3 Use the shuttle sort to sort the list 6, 4, 8, 2, 4, 13, 5 into ascending order. Show the list at the end of each pass of the algorithm, and record the numbers of comparisons and swaps involved in each pass of the algorithm.

4 Two different algorithms are being used to process a set of n objects. Algorithm A has order n^2 and algorithm B has order $n \log n$. When Algorithm A is used on a set of 500 objects it takes 2.4 seconds to complete. When Algorithm B is used on the same set of 500 objects it takes 10.7 seconds to complete. Roughly how long will each algorithm take when used on a set of:

- (a) 2000 objects?
- (b) 10,000 objects?

Commented [NS(1): After the first pass $A=16, B=36$
After the second pass $A=36, B=20$
After the third pass $A=16, B=4$
After the fourth pass $A=12, B=4$
After the fifth pass $A=8, B=4$
Step 6 is not encountered again, and the output is 4.

Commented [NS(2): No times. A changes to 20, then B changes to 20 then we stop at Step 5.

Commented [NS(3): Two fewer than the maximum of A and B . Because if $B=1$ then the value of A reduces by one in each pass, so it will take $A-1$ passes to reduce A to the value 1. Step 6 is encountered in the last pass, so Step 6 will be encountered $A-2$ times.

Commented [NS(4): If $A=1, B=0$, for example, the algorithm would never terminate.

Commented [NS(5): A
B, C, D
E, F
G, H
I, J

Commented [NS(6): A, C, D
B, E
F, G, J
H
I

Commented [NS(7): Decreasing order $A(12.4), H(8.8), B(8.1), I(7.5), F(7.4), E(7.3), J(3.2), C(2.3), G(1.7), D(1.2)$. Gives:

A, J
H, C, G, D
B, I
F, E

Note that 4 boxes is optimal, because the total size divided by 16 is 3.74.

Commented [NS(8): 6, 4, 8, 2, 4, 13, 5
4, 6, 8, 2, 4, 13, 5 (1 comparison, 1 swap)
4, 6, 8, 2, 4, 13, 5 (1 comparison, 0 swaps)
2, 4, 6, 8, 4, 13, 5 (3 comparisons, 3 swaps)
2, 4, 4, 6, 8, 13, 5 (3 comparisons, 2 swaps)
2, 4, 4, 6, 8, 13, 5 (1 comparison, 0 swaps)
2, 4, 4, 5, 6, 8, 13 (4 comparisons, 3 swaps)

Commented [NS(9): A: $2.4 * 2000^2 / 500^2 = 38.4$ seconds
B: $10.7 * 2000 \log 2000 / 500 \log 500 = 523$ seconds.

Commented [NS(10): A: $2.4 * 10000^2 / 500^2 = 960$ seconds (16 minutes)
B: $10.7 * 10000 \log(10000) / 500 \log(500) = 317$ seconds (5 minutes 17 seconds)

Commented [NS(11R10): The moral of the story is that algorithms with lower time complexity will always *eventually* be quicker

5 Tamsin is sorting a list of numbers into increasing order. After the second pass of her algorithm she has the following list:

8 6 1 10 5 6 12

(a) Explain how you know that Tamsin is not using the Shuttle Sort.

In fact, Tamsin is using the Bubble Sort.

(b) Find all the possibilities for Tamsin’s original list.

(c) How many comparisons in total are needed to sort Tamsin’s list?

6 The following 16 items need to be packed into boxes which have maximum capacity of 10 kg.

Item	A	B	C	D	E	F	G	H
Mass (kg)	6.4	1.5	8.1	5.0	2.2	7.4	1.8	8.7
Item	I	J	K	L	M	N	O	P
Mass (kg)	2.4	2.8	3.6	2.5	3.6	6.9	1.9	6.7

(a) Use the full bins method, followed by the first-fit decreasing algorithm to find a possible packing.

(b) Explain how you know that the packing you found is optimal.

7 Wilfred is sorting a list of numbers into increasing order. After the third pass of his algorithm he has the following list:

2 5 7 10 3 9 12

(a) Explain how you know that Wilfred is not using a bubble sort.

In fact, Wilfred is using the shuttle sort.

(b) How many possibilities are there for the Wilfred’s original list?

(c) What are the maximum and minimum numbers of comparisons that Wilfred has done so far?

Commented [NS(12): The first two elements are out of order

Commented [NS(13): Because the 8 and 6 are out of order, something larger must have been before or between them. The only possibilities are the 10 and the 12. If it is the 12 then nothing else will change relative ordering, and the possibilities are
12, 8, 6, 1, 10, 5, 6 or 8, 12, 6, 1, 10, 5, 6.

If the 10 is before or between the 8 and 6 then it must meet the 12 at its current position in the list, so the possibilities are

10, 8, 6, 1, 12, 5, 6 or 8, 10, 6, 1, 12, 5, 6.

Commented [NS(14): You probably need to play it through to decide whether the bubble sort will terminate early:

6, 1, 8, 5, 6, 10, 12

1, 6, 5, 6, 8, 10, 12

1, 5, 6, 6, 8, 10, 12

So it does terminate early, but only by one pass. So the total number of comparisons are $6+5+4+3+2 = 20$

Commented [NS(15): I call this the full bins *method* because there is no algorithm as such to find the full bins. I find it useful to write the elements in decreasing order to find the full bins:

H(8.7)

C(8.1)

F(7.4)

N(6.9)

P(6.7)

A(6.4)

D(5.0)

K(3.6)

M(3.6)

J(2.8)

L(2.5)

I(2.4)

E(2.2)

O(1.9)

G(1.8)

B(1.5)

This gives me full bins of {C, O}, {P, G, B}, {A, K}, {D, J, E}. The remaining 6 items then pack as follows:

H

F, L

N, I

M

Commented [NS(16): We used 8 boxes. 7 boxes have a maximum capacity of 70 kg, and the total weight of the items is 71.5 kg.

Commented [NS(17): The 10 should be second from the end after 2 passes.

Commented [NS(18): $3!=24$

Commented [NS(19): If the first four numbers started in the correct order, then just 3 comparisons will be needed. But if they were in the exact opposite order then $1+2+3=6$ comparisons would be required.

8 The following instructions operate on positive integers greater than 4.

- STEP 10 Choose any positive integer greater than 4, and call it n .
- STEP 15 Write down n .
- STEP 20 If n is even then let $n = \frac{n}{2}$ and write down the result
- STEP 30 If n is odd then let $n = 3n + 1$ and write down the result.
- STEP 40 Go to Step 20.

- (a) Apply the instructions with 6 as the chosen integer, stopping when a sequence repeats itself.
- (b) Apply the instructions with 256 as the chosen integer, stopping when a sequence repeats itself.
- (c) Add an instruction to stop the process when n becomes equal to 1.
- (d) It is not known if, when modified as in Part (c), the instructions form an algorithm. What would need to be known for it to be an algorithm?

9 Lists of numbers are created by starting with the numbers 1 to n in order. The number 1 is then swapped at random with one of the numbers 2 to n .

- (a) What is the maximum and minimum number of comparisons that will be needed when sorting lists of this type using the Shuttle sort?
- (b) Hence deduce the average time complexity, in terms of n , of the shuttle sort when applied to lists of this type.

10 Lists of numbers are created by starting with the numbers 1 to n in order. The number 1 is then swapped at random with one of the numbers 2 to n .

- (a) If the number 1 is swapped with the number k , for $k = 2, 3, \dots, n$, then how many passes of the bubble sort algorithm will be required to sort the resulting list? How many comparisons will be required?
- (b) Hence deduce the average time complexity, in terms of n , of the bubble sort when applied to lists of this type.

Commented [NS(20): You get the sequence 6, 3, 10, 5, 16, 8, 4, 2, 1, 4 at which point we have a repeat.

Commented [NS(21): You get the sequence 256, 128, 64, 32, 16, 8, 4, 2, 1, 4 at which point we have a repeat.

Commented [NS(22): STEP 35: If $n=1$ then STOP

Commented [NS(23): Every algorithm must terminate, so we need to know if every starting value eventually leads to and output of 1.

This is unknown, and a famous mathematical problem known as the Collatz conjecture. See https://en.wikipedia.org/wiki/Collatz_conjecture and <https://www.youtube.com/watch?v=094y1Z2wpJg>

Commented [NS(24): If the 1 is swapped with the 2 we will need $n-1$ comparisons (the same as if the list was sorted). If the 1 is swapped with the n we will need $n-2 + n-1 = 2n-3$ comparisons

Commented [NS(25): Both best and worst case are linear, so on average what is happening here is $O(n)$

Commented [NS(26): It will take $k-1$ passes to get the 1 back to the start, at which point the list must be in the correct order. It will then require one more pass to discover that the list is correctly sorted. So k passes in total UNLESS $k=n$ in which case we need $n-1$ passes (because the list reaches length 1).

The total number of comparisons for k passes is $(n-1)+(n-2)+\dots+(n-k) = kn - \frac{1}{2}k(k+1) = \frac{1}{2}k(2n+k+1)$

Commented [NS(27): The average number of comparisons is found by summing the above expression from $k=2$ to $n-1$, adding a second expression when $k=n-1$ and then dividing the total by $n-1$. The numerator is clearly going to be cubic in n , because it is the sum of roughly n quadratic terms, and the denominator is linear. Therefore the algorithm has quadratic order on average. $O(n^2)$